

Society Management System

Enterprise Laravel Finance & Accounting Module — Implementation Plan

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Laravel Finance Module Package - Implementation Plan

1. Executive Summary

This document outlines the end-to-end technical plan for designing, developing, and integrating a modular, enterprise-grade **Finance & Accounting Package** (`Modules/Finance`) for the Society Management System.

The package will operate as a self-contained, pluggable Laravel module with its own service provider, database migrations, models, services, controllers, routes, data tables, blade views, and automated tests. It delivers a comprehensive **Double-Entry Accounting Engine, Automated Maintenance Billing, Accounts Payable / Vendor Vouchers, Banking & BRS, Sinking / Reserve Funds, and Financial Reporting & Defaulter Tracking.**

2. Package Architecture & Folder Structure

The module resides in `Modules/Finance` and is autoloaded via PSR-4 in `composer.json`:

```
"autoload": {
    "psr-4": {
        "App\\": "app/",
        "Modules\\Finance\\": "Modules/Finance/src/"
    }
}
```

```
c:\laragon\www\SMP\society-management\Modules\Finance\
??? config/
?   ??? finance.php                # Module configuration (currency, default accounts, tax settings)
??? database/
?   ??? migrations/                # Dedicated finance migrations
?   ?   ??? 2026_09_01_000001_create_finance_chart_of_accounts_table.php
?   ?   ??? 2026_09_01_000002_create_finance_fiscal_years_table.php
?   ?   ??? 2026_09_01_000003_create_finance_journal_entries_tables.php
?   ?   ??? 2026_09_01_000004_create_finance_bank_accounts_tables.php
?   ?   ??? 2026_09_01_000005_create_finance_vendors_and_bills_tables.php
?   ?   ??? 2026_09_01_000006_create_finance_payment_vouchers_table.php
?   ?   ??? 2026_09_01_000007_create_finance_petty_cash_table.php
?   ?   ??? 2026_09_01_000008_create_finance_invoices_and_items_tables.php
?   ?   ??? 2026_09_01_000009_create_finance_payments_table.php
?   ?   ??? 2026_09_01_000010_create_finance_funds_and_fds_table.php
?   ?   ??? 2026_09_01_000011_create_finance_audit_logs_table.php
?   ??? seeders/
?       ??? FinanceDatabaseSeeder.php
?       ??? ChartOfAccountsSeeder.php    # Standard Society COA
?       ??? FinancePermissionsSeeder.php # Spatie permissions for finance
??? resources/
?   ??? views/                      # Blade UI (finance:: prefix)
?   ?   ??? dashboard/              # Financial dashboard & KPIs
?   ?   ??? accounts/               # Chart of accounts tree & manual journals
?   ?   ??? billing/                # Invoices, batch generator, invoice templates
?   ?   ??? payments/               # Inward receipts, passbook, receipt PDFs
?   ?   ??? payables/               # Vendors, vendor bills, vouchers
?   ?   ??? petty-cash/              # Daily cash register
?   ?   ??? banking/                # Bank accounts & BRS reconciliation
?   ?   ??? funds/                  # Sinking/Reserve funds & FDs
?   ?   ??? reports/                # P&L, Balance Sheet, Trial Balance, Aging
?   ??? pdf/
?       ??? invoice.blade.php        # Downloadable PDF invoice template
```

```

??? DataTables/                                # Yajra DataTables integration
?   ??? AccountsDataTable.php
?   ??? InvoicesDataTable.php
?   ??? PaymentsDataTable.php
?   ??? VendorBillsDataTable.php
?   ??? VouchersDataTable.php
?   ??? BankTransactionsDataTable.php
??? Events/                                    # Domain Events (InvoiceGenerated, PaymentReceived, etc.)
??? Http/
?   ??? Controllers/
?   ?   ??? FinanceDashboardController.php
?   ?   ??? AccountController.php
?   ?   ??? JournalEntryController.php
?   ?   ??? InvoiceController.php
?   ?   ??? PaymentController.php
?   ?   ??? VendorController.php
?   ?   ??? VendorBillController.php
?   ?   ??? PaymentVoucherController.php
?   ?   ??? PettyCashController.php
?   ?   ??? BankAccountController.php
?   ?   ??? BankReconciliationController.php
?   ?   ??? FundController.php
?   ?   ??? FinancialReportController.php
?   ??? Requests/                             # Form Requests / Validation rules
??? Models/                                   # Eloquent Models
?   ??? Account.php
?   ??? FiscalYear.php
?   ??? JournalEntry.php
?   ??? JournalItem.php
?   ??? BankAccount.php
?   ??? BankTransaction.php
?   ??? Vendor.php
?   ??? VendorBill.php
?   ??? PaymentVoucher.php
?   ??? PettyCashEntry.php
?   ??? Invoice.php
?   ??? InvoiceItem.php
?   ??? Payment.php
?   ??? Fund.php
?   ??? FinancialAuditLog.php
??? Services/                                 # Core Business Logic Layer
?   ??? AccountingEngineService.php           # Double-entry ledger postings
?   ??? BillingGenerationService.php           # Batch invoice calculation engine
?   ??? AdvancePaymentService.php             # Prepaid wallet allocation
?   ??? BankReconciliationService.php          # BRS statement matching
?   ??? VoucherApprovalService.php            # Workflow approval state machine
?   ??? FinancialReportService.php            # Financial reports builder
?   ??? LegacyDataMigrationService.php        # Port legacy tables into new ledger
??? FinanceServiceProvider.php                # Main package service provider

```

3. Database Architecture & Relationships

```

erDiagram
    Account ||--o{ JournalItem : "contains"
    JournalEntry ||--|{ JournalItem : "has lines"
    FiscalYear ||--o{ JournalEntry : "belongs to"

    User ||--o{ Invoice : "billed to"
    Flat ||--o{ Invoice : "billed for"
    Invoice ||--|{ InvoiceItem : "has line items"
    Invoice ||--o{ Payment : "settled by"

    Vendor ||--o{ VendorBill : "submits"
    VendorBill ||--o{ PaymentVoucher : "paid via"
    BankAccount ||--o{ PaymentVoucher : "disburses"
    BankAccount ||--o{ Payment : "receives"
    BankAccount ||--o{ BankTransaction : "reconciles"

    JournalEntry ||--o{ Invoice : "linked transaction"
    JournalEntry ||--o{ Payment : "linked transaction"
    JournalEntry ||--o{ PaymentVoucher : "linked transaction"

```

4. Core Functional Modules

4.1. General Ledger & Double-Entry Accounting Engine

- **Chart of Accounts (COA):** Pre-configured hierarchy for housing societies:
 - **Assets:** Cash on Hand, Bank Accounts, Fixed Deposits, Accounts Receivable (Member Dues).
 - **Liabilities:** Security Deposits, Advance Maintenance Received, Accounts Payable (Vendors).
 - **Equity / Funds:** General Society Corpus, Sinking Fund, Major Repair & Renovation Reserve.
 - **Income:** Maintenance Fees, Sinking Fund Charges, Late Payment Fines, NOC/Transfer Fees, Facility Booking Fees, Bank Interest.
 - **Expenses:** Security Service, Housekeeping, Lift AMC, Electricity & Water, Gardening, Repair & Maintenance, Administrative Expenses.
- **Automated Journal Entries:** Every transaction (bill generation, member payment, vendor bill, voucher payout) automatically generates a balanced Debit/Credit journal record without manual entry required.
- **Manual Adjustments:** Supports manual journal entries for depreciation, interest accrual, bad debt write-offs, and year-end closing.

4.2. Accounts Receivable (AR) & Automated Invoicing

- **Multi-Method Billing Engine:**
 - Fixed flat rate (e.g. ₹2,500/month).
 - Category-based (1 BHK, 2 BHK, 3 BHK, Villa, Tenement).
 - Area-based rate (₹/sq.ft).
 - Utility/Metered add-ons (Water, DG Power, Parking Slot).
- **Batch Bill Generation:** One-click generation for all occupied flats for a target month/quarter with automated due date and late penalty rules.
- **Auxiliary Invoices:** Instant creation of one-off invoices for Name Transfer, NOC, Clubhouse/Hall Booking, Tenant Move-in/Move-out charges.
- **Prepaid Maintenance Adjustments:** Automatically deducts charges from resident prepaid balances before computing net payable.
- **Digital Invoices & Receipts:** Automated branded PDF invoice generation and instant receipt printing.

4.3. Accounts Payable (AP), Procurement & Petty Cash

- **Vendor Master:** Database of contractors/suppliers with GSTIN, PAN, Bank Details, and service contracts (AMC).
- **Vendor Invoices & Work Orders:** Record incoming vendor bills with expense classification and due dates.
- **Multi-Tier Payment Vouchers:**
 - Creator: Accountant / Secretary draft voucher.
 - Approval: Treasurer / Committee review and approve.
 - Disbursement: Bank transfer / Cheque / Cash payout with automated journal posting.

- **Petty Cash Book (Imprest System):** Daily petty cash disbursement tracking, digital vouchers, balance threshold alerts, and replenishment requests.

4.4. Banking, Treasury & Bank Reconciliation Statement (BRS)

- **Multi-Account Management:** Support for multiple Society Bank Accounts (Current, Savings, Escrow) and Petty Cash registers.
- **Bank Reconciliation Statement (BRS):**
 - Upload bank statements (CSV/Excel).
 - Automated matching of bank credits against member receipts and bank debits against vouchers.
 - Identification of uncredited cheques, uncleared deposits, and bank charges.
- **Fund & Fixed Deposit (FD) Management:**
 - Sinking Fund and Reserve Fund ledger tracking.
 - FD certificates, maturity dates, interest reinvestment, and accrual schedule.

4.5. Financial Reporting & Analytics

- **Standard Financial Statements:**
 - **Income & Expenditure Account (Profit & Loss)**
 - **Balance Sheet** (Assets vs. Liabilities & Funds)
 - **Trial Balance** (Period-to-date and year-to-date)
 - **General Ledger & Day Book**
- **Society-Specific Reports:**
 - **Member Passbook (Statement of Account):** Real-time ledger view per flat/resident.
 - **Dues Aging Report:** Classification of arrears into 0-30 , 31-60 , 61-90 , and 90+ days.
 - **Defaulters Tracker:** List of overdue members with automated reminder notices.
 - **Collection Efficiency & Expense Budget vs. Actual Analysis.**
- **Export Formats:** All reports exportable in high-quality Excel (`.xlsx`) and PDF formats.

5. Security & Role-Based Access Control (RBAC)

The module defines granular Spatie permissions:

```
finance_dashboard_view
finance_invoice_view
finance_invoice_create
finance_invoice_cancel
finance_payment_receive
finance_receipt_print
finance_vendor_manage
finance_bill_create
finance_voucher_create
finance_voucher_approve
finance_voucher_pay
finance_petty_cash_manage
finance_bank_manage
finance_bank_reconcile
```

```
finance_journal_create
finance_coa_manage
finance_reports_view
finance_reports_export
finance_settings_manage
```

6. Phased Implementation Roadmap

```
????????????????????????????????????????????????????????????????????????????????????
? PHASE 1: Package Skeleton & Core Accounting Engine ?
? ??? Register Modules/Finance namespace in composer.json ?
? ??? Build FinanceServiceProvider & configuration files ?
? ??? Migrations: Chart of Accounts, Fiscal Years, Journal Entries, Bank Accts?
? ??? Seed standard Society Chart of Accounts & Permissions ?
? ??? Implement AccountingEngineService (Debit/Credit balance engine) ?
????????????????????????????????????????????????????????????????????????????????????
?
?
????????????????????????????????????????????????????????????????????????????????????
? PHASE 2: Invoicing & Accounts Receivable (AR) ?
? ??? Migrations: Invoices, Invoice Items, Payments ?
? ??? Implement BillingGenerationService (Batch billing & rate algorithms) ?
? ??? Build Invoices & Payments Controllers, DataTables, and UI ?
? ??? Generate downloadable PDF Invoices & Official Payment Receipts ?
? ??? Resident Statement of Account (Member Passbook) ?
????????????????????????????????????????????????????????????????????????????????????
?
?
????????????????????????????????????????????????????????????????????????????????????
? PHASE 3: Accounts Payable (AP), Vendors & Petty Cash ?
? ??? Migrations: Vendors, Vendor Bills, Payment Vouchers, Petty Cash ?
? ??? Vendor master directory & contract management ?
? ??? Multi-tier Payment Voucher approval workflow (Draft -> Approved -> Paid)?
? ??? Petty Cash Imprest tracking & replenishment ?
? ??? Automatic Journal posting for all payables & disbursements ?
????????????????????????????????????????????????????????????????????????????????????
?
?
????????????????????????????????????????????????????????????????????????????????????
? PHASE 4: Banking, BRS & Fund Management ?
? ??? Multi-bank account management & Cash register ?
? ??? Bank Reconciliation Statement (BRS) tool with statement upload ?
? ??? Sinking Fund, Reserve Fund & Fixed Deposit (FD) Ledger ?
? ??? FD maturity and interest tracking ?
????????????????????????????????????????????????????????????????????????????????????
?
?
????????????????????????????????????????????????????????????????????????????????????
? PHASE 5: Reports, Legacy Data Migration & Verification ?
? ??? Financial Statements: P&L, Balance Sheet, Trial Balance, Daybook ?
? ??? Defaulters List & Aging Report (0-30, 31-60, 61-90, 90+ days) ?
? ??? Build LegacyDataMigrationService to import existing maintenance/expenses?
? ??? Comprehensive PHPUnit test suite (Feature & Unit tests) ?
? ??? UI navigation links integration with the main sidebar ?
????????????????????????????????????????????????????????????????????????????????????
```

7. Data Migration Strategy (Legacy Synchronization)

To ensure zero data loss from existing tables (`expenses` , `expense_categories` , `maintenances` , `maintenance_bills` , `prepaid_maintenances` , `name_transfer_bills`):

1. **Migration Command:** `php artisan finance:migrate-legacy`

2. **Action:**

- Maps each `expense_category` to a corresponding Account head in Chart of Accounts.
- Converts `expenses` into `VendorBills` / `PaymentVouchers` with journal entries.

- Converts `maintenance_bills` and `maintenances` into `Invoices` and `Payments` .
- Populates member opening balances and reconciles historical totals.

8. Verification & Testing Checklist

- ☐ **PSR-4 Autoloading:** Module classes load seamlessly via `composer dump-autoload` .
- ☐ **Migration Check:** `php artisan migrate` creates all finance tables with foreign key constraints.
- ☐ **Accounting Balance Integrity:** Test that every generated transaction has `sum(debit) == sum(credit)` .
- ☐ **Batch Invoicing:** Run monthly batch bill generation for 100+ flats within 2 seconds.
- ☐ **Payment Workflow:** Partial and full payments correctly update invoice status to `paid` and reflect on resident passbook.
- ☐ **Voucher Approval:** Non-authorized users cannot disburse funds without committee approval.
- ☐ **Report Accuracy:** Balance sheet equation holds true (`Assets = Liabilities + Equity`).
- ☐ **Export Verification:** Excel and PDF outputs render with proper headers, currency symbols, and accurate calculations.